Dingli Liang

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PHD STUDY AREA

Algebraic Number Theory/ Arithmetic Algebraic Geometry/ Iwasawa Theory over Function Field/ Non-Noetherian Iwasawa Theory/ Integral Iwasawa Theory/ Non-commutative Iwasawa Theory/ p-adic Hodge Theory/ Algebraic K-Theory

AI RESEARCH AREA

Post-training/Reinforcement Learning/Agentic RL/AI Safety/AI for Math/Geometric Deep Learning

PUBLICATIONS

Arithmetic Geometry

• David Burns, Alexandre Daoud and Dingli Liang, *On non-Noetherian Iwasawa Theory*, accepted by *Compositio Mathematica*

• David Burns, Kuang Yu, and Dingli Liang, *On The Coherency of Completed Group Algebras*, *Journal of Algebra*, 661(2025), 757-777

• Dingli Liang and Meng Fai Lim, *On the Iwasawa Asymptotic Class Number Formula for $\mathbb{Z}_p^r \rtimes \mathbb{Z}_p$-extensions*, *Acta. Arithmetica*, 189 (2019), no. 2, 191-208

• On-going Research: On Non-Commutative Supersingular Iwasawa Theory, with David Burns

Artificial Intelligence (First/Co-First Author)

• *CapMem: A Benchmark for Caption-Based Episodic Memory in Egocentric Video*, first author., EMNLP 2026

• *Structure-Preserving Visual Token Compression for Streaming Video Understanding via Bilateral Grid Splatting*, co-first author, submitted to iCLR 2027

• *Hölder Policy Optimisation*, co-first author, submitted to iCLR 2027, arXiv 2605.12058v1

• *StoReD: Stochastic Readout of Distributional Memory*, co-first author, submitted to ARR Oct 2026

• *WorldCupArena: Fine-Grained Evaluation of Language Models and Deep-Research Agents on Football Forecasting*, final author, submitted to ICLR 2027, arxiv 2607.18084

• Ongoing Research: *Recursive Self-Improvement via Adaptive Decoding-Time Exploitation*, first author, with Zhiyuan He

• Ongoing Research: *Regularized Reinforcement Learning for Bayesian Posterior Forests*, co-first author with Zhiyuan He

• Ongoing Research: *ChronoHarness: Recursive Self-Improvement of Forecasting Agents through Time-Consistent Historical Replay*, first author, with Jinyang Zhang

AI Work EXPERIENCE

Huawei London Research Center *Internship, AI research* 07/2025– 02/2026

Duties: Collect cutting-edge academic and industry developments, provide strategic suggestion for technologies plan

• Focusing on LLM/MLLM post-training, including SFT-then-RL paradigm, and LLM-based reasoning

• Researching RL algorithms (GRPO, GSPO, DAPO, GMPO)

• Researching how diffusion language models accelerate LLM inference and improve reasoning capabilities

• Exploring cutting-edge AI for Math/Math agent techniques, like automated problem discovery systems and theorem provers.

• Improving multimodal reasoning and streaming video understanding.

EDUCATION

Imperial College London/University College London London, UK

PhD Student/ Study in London School of Geometry and Number Theory/ Supervised by Prof. David Burns 09/2021 –

King's College London London, UK

Visiting Research Student, Mathematics 09/2020–11/2025

Wuhan University Wuhan, China

First Class Honours Bachelor Degree in Science, Mathematics and Applied Mathematics GPA: 3.51/4.0 09/2014–06/2018

Peking University Beijing, China

Visiting Student 02/2018–05/2018

AWARDS, PRIZES & SCHOLARSHIPS

Engineering and Physical Sciences Research Council (EPSRC) CDT PhD Full-funding 09/2020–09/2024

Outstanding Graduates of Hongyi Honor College, Wuhan University 06/2018

B/C Level Scholarship for Triple A Student, Wuhan University 10/2015 10/2016

WORK EXPERIENCE

Imperial College London *Graduate Teaching Assistant, Mathematics Department* 09/2023–

• Statistical Modeling • Supervised Learning(Postgraduate Level) • Homological Algebra(Research Postgraduate Level)

• Linear Algebra and Numerical Analysis

University College London *Graduate Teaching Assistant, Mathematics Department* 09/2023–05/2024
• Galois Theory(Postgraduate Level) • Linear Algebra

King's College London *Graduate Teaching Assistant, Mathematics Department* 09/2022–12/2022
• Sequence and Series

London School of Economics and Political Science *Guest Teacher, Mathematics Department* 09/2024–12/2024
• Quantitative method

The Academy of Mathematics and Systems Science of the Chinese Academy of Sciences Beijing,China
Visiting Scholar, Morningside Center of Mathematics 10/2020–01/2022

COMPETITION EXPERIENCE

Mathematical Contest in Modeling(MCM), SP Prize	Undergraduate	01/2016
The Chinese Mathematics Competitions, Second Prize	Undergraduate	10/2015
Chinese Physics Olympiad, Second Prize	High School	09/2012, 09/2013
China Adolescent Robotics Competition, First Prize	Middle School	06/2009

CORE SKILLS

• **Programming:** Python, C++, C, Lean, Pascal, SQL • **Algorithms:** Data Structures
• **Deep Learning & NLP:** PyTorch, Hugging Face (Transformers, Datasets), vLLM
• **Machine Learning & Data:** NumPy, SciPy, scikit-learn, Pandas, Dask, Matplotlib, Seaborn
• **AI Agents & Engineering:** LangChain, Langgraph, CrewAI, OpenClaw, Asyncio
• **GPU Training & Systems:** Linux/Ubuntu, Docker, Slurm, Tmux • **Research:** LaTeX, Markdown

PROJECTS

Agentic-RL Lab Open-Source Contribution — GiGPO for ALFWorld
https://github.com/datawhalechina/agentic-rl-lab
• Implementation: Developed a GiGPO training pipeline on PyTRIO and ALFWorld, integrating trajectory-level and observation-grouped step-level advantage estimation. Unified GRPO and GiGPO training within a single framework.
• Experimental study: Conducted controlled evaluations under consistent model, LoRA, reward, sampling, and data-order settings. Allocated a fixed budget of 80 steps and 5,120 trajectories per experiment.
• Infrastructure: Built workflows for data processing, model training, checkpoint evaluation, and result analysis. Implemented metrics tracking for success rate, invalid actions, and token consumption, together with confidence interval estimation.
• Quality assurance: Developed test suites for advantage calculation, token-logprob alignment, and training/evaluation pipelines. Contributed documentation covering algorithm design, usage, and environment setup.

LLM-Driven Dispatch and Operations Agent for Chauffeur Services
• Developed a router-based workflow and state-machine orchestration layer to coordinate dispatching, customer support, and internal business operations.
• Built a constrained ReAct agent loop leveraging native tool integrations and rule-based validation for reliable task routing.
• Designed and implemented an agent runtime framework with secure tool invocation, environment sandboxing, Human-in-the-Loop review, checkpoint recovery, and execution tracing.
• Ensured execution reliability through database transactions, idempotent operations, and restart recovery.
• Implemented version-controlled memory management with scoped contexts and human-validated feedback incorporation.
• Created a simulation and evaluation platform for continuous agent optimization, supporting user-request generation, conversation replay, human feedback collection, and automated regression testing.

Solana New-Token Research and Trading Agent
• Built a real-time market data pipeline for Pump.fun and Raydium LaunchLab, monitoring token launches, bonding-curve activity, and liquidity migrations with robust validation, deduplication, reconnection, and observability mechanisms.
• Designed a token evaluation framework that ranked opportunities based on trading momentum, wallet behavior, holder concentration, and permission analysis, balancing opportunity potential, smart-money signals, and risk factors.
• Developed an automated due-diligence workflow using DeepSeek Search and Fetch to verify project information from token metadata, websites, and public social content, aggregating supporting and contradictory evidence with source attribution.
• Implemented a high-fidelity trading simulator for bonding-curve markets, modeling execution latency, fees, liquidity constraints, slippage, and market impact to evaluate trading strategies under realistic conditions.
• Built a paper-trading and evaluation platform to track portfolio performance, positions, and P&L while maintaining persistent records of observations, decisions, and outcomes for systematic analysis and offline benchmarking.